Chapter 15. The Probabilistic Method

Who Knows What It Looks Like, But It Exists

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15.1 The notion of probability

Motivation. On a finite sample space with equally likely outcomes, probability is counting divided by counting. The union bound $P(\bigcup A_i) \le \sum P(A_i)$, trivial as it is, will drive the existence proofs of the next section.

Tossing a coin four times has $2^4 = 16$ outcomes, five of which ($HHHH$, $HHHT$, $HHTH$, $HTHH$, $THHH$) have at least three heads, so this happens with probability $5/16$.

Definition 15.1 (Probability)

Let Ω be a finite set of equally likely outcomes (the **sample space**) and $A \subseteq \Omega$ an **event**. The **probability** of A is $P(A) = \frac{|A|}{|\Omega|}$.

For disjoint A, B, $P(A \cup B) = P(A) + P(B)$; in general $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ by the sieve. The **complement** of A is $\bar{A} = \Omega - A$, and $P(\bar{A}) = 1 - P(A)$.

Proposition 15.2 (Union bound)

For events $A_1, \dots, A_n$, $P(A_1 \cup \cdots \cup A_n) \le P(A_1) + \cdots + P(A_n)$.

Proof. $|A_1 \cup \cdots \cup A_n| \le |A_1| + \cdots + |A_n|$: the left side counts every element of the union once, the right side at least once. ∎

Example 15.3

Six numbers are drawn from $[36]$. A ticket of six numbers contains at least one winning number with probability

$$1 - \binom{30}{6} \Big/ \binom{36}{6} \approx 1 - 0.3048 = 0.6952,$$

not $\frac{6}{36}$.

Example 15.4 (Birthday paradox)

Among 23 people (nobody born on February 29), two share a birthday with probability more than $\frac{1}{2}$. The probability that all birthdays differ is

$$\frac{365 \cdot 364 \cdots 343}{365^{23}} < \frac{1}{2},$$

and more people only increase the chance of a coincidence.

15.2 Non-constructive proofs

Motivation. If a random object has a bad property with probability less than 1, some object avoids it. This proves that colourings without large monochromatic cliques exist, giving $R(k,k) > 2^{k/2}$, although nobody knows how to construct such colourings explicitly.

Theorem 15.5 (Erdős)

$R(k,k) > 2^{k/2}$ for $k \geq 3$.

Proof. Let $n \leq 2^{k/2}$ and colour each edge of K_n red or blue by an independent fair coin; Ω is the set of all $2^{\binom{n}{2}}$ colourings, equally likely. A fixed K_k is monochromatic in 2 of the $2^{\binom{k}{2}}$ colourings of its edges, so with probability $2^{1-\binom{k}{2}}$. Let A_S be the event that the K_k-subgraph S is monochromatic. By Proposition 15.2,

$$P\Big(\bigcup_S A_S\Big) \leq \sum_S P(A_S) = \binom{n}{k} 2^{1-\binom{k}{2}}. \tag{15.1}$$

Using $\binom{n}{k} < \frac{n^k}{k!}$ and $n^k \leq 2^{k^2/2}$,

$$\binom{n}{k} 2^{1-\binom{k}{2}} < \frac{2^{k^2/2}}{k!} \cdot 2^{1-\frac{k^2}{2}+\frac{k}{2}} = \frac{2 \cdot 2^{k/2}}{k!} < 1$$

for $k \geq 3$: at $k = 3$ the value is $\frac{2 \cdot 2^{3/2}}{6} < 1$, and increasing k by one multiplies it by $\frac{\sqrt{2}}{k+1} < 1$. So with positive probability no K_k is monochromatic, and such a colouring of K_n exists. ∎

Together with Corollary 13.8, $(\sqrt{2})^k < R(k,k) \leq 4^{k-1}$; these are essentially the best known exponential bounds, apart from the recent $(4-\epsilon)^k$ upper bound.

Theorem 15.6

Let $n, m \geq 2$ with $m \geq 2\log_2 n$. Then the edges of $K_{n,n}$ can be 2-coloured with no monochromatic $K_{m,m}$.

Proof. Colour randomly. A fixed $K_{m,m}$ (choose m vertices on each side: $\binom{n}{m}^2$ choices) is monochromatic with probability $2 \cdot 2^{-m^2}$. By the union bound it suffices that $2\binom{n}{m}^2 < 2^{m^2}$. Indeed, $\binom{n}{m} \leq \frac{n^m}{m!}$ and $m! \geq 2$ give $2\binom{n}{m}^2 < n^{2m}$, and $n \leq 2^{m/2}$ gives $n^{2m} \leq 2^{m^2}$. ∎

Equivalently, there is an $n \times n$ 0–1 matrix with no constant $m \times m$ submatrix for $m \approx 2\log_2 n$; the best explicit constructions only achieve $m = c\sqrt{n}$.

15.3 Independent events

Motivation. Sometimes $P(A \cap B) = P(A)P(B)$ and sometimes not; independence means that learning B does not change the odds of A. Conditional probabilities defy intuition (Simpson's paradox, Monty Hall), and the reliable tool is the law of total probability, called Bayes' theorem in the book.

15.3.1 The notion of independence and Bayes' theorem

For a number chosen from [12], divisibility by 2 and by 3 satisfy the product rule, but divisibility by 2 and by 4 do not: knowing evenness halves the outcomes without losing any multiple of 4.

Definition 15.7 (Independence)

Events A, B are **independent** if $P(A \cap B) = P(A)P(B)$, and **dependent** otherwise.

Definition 15.8 (Conditional probability)

For $P(B) > 0$, $P(A \mid B) = \frac{P(A \cap B)}{P(B)}$ is the probability of A given B.

Proposition 15.9

For $P(B) > 0$, A and B are independent iff $P(A \mid B) = P(A)$.

Proof. Divide $P(A \cap B) = P(A)P(B)$ by $P(B)$, or multiply back. ■

Example 15.10

Four coin tosses, at least two heads (B). The probability of four heads is $P(A \mid B) = \frac{P(A)}{P(B)} = \frac{1/16}{11/16} = \frac{1}{11}$ (with $P(B) = \frac{1+4+6}{16}$).

Example 15.11

For a random n-permutation let $A = \{p_1 > p_2\}$ and $B = \{p_2 > p_3\}$. Then $P(A) = P(B) = \frac{1}{2}$ (reverse the pair) and $P(A \cap B) = P(p_1 > p_2 > p_3) = \frac{1}{6}$. So $P(A \mid B) = \frac{1}{3} \neq P(A)$, and the events are dependent.

Example 15.12 (Simpson's paradox)

Two colleges each admit domestic applicants at a higher rate than international ones. It does not follow that the whole university does:

	Liberal Arts	Engineering	University
Domestic	10/120 (8.3%)	10/10 (100%)	20/130 (15.4%)
International	1/15 (6.7%)	90/100 (90%)	91/115 (79.1%)

About 87% of the international applicants, but only 8% of the domestic ones, applied to Engineering, where admission is easy.

Theorem 15.13 (Bayes' theorem, law of total probability)

If A, B are mutually exclusive with $A \cup B = \Omega$ and C is any event, then

$$P(C) = P(C \mid A)P(A) + P(C \mid B)P(B). \tag{15.2}$$

Proof. C is the disjoint union of $C \cap A$ and $C \cap B$, so $P(C) = P(C \cap A) + P(C \cap B)$, and $P(C \cap A) = P(C \mid A)P(A)$, likewise for B. ■

So $P(C)$ is a *weighted* average of the conditional probabilities. In Example 15.12 the weights (the shares applying to each college) differ between the two groups, which explains the paradox. With equal weights the domestic applicants would win.

Example 15.14

Exactly one of Andrew (A) and Brandi (B) committed a crime; the culprit has a blood type

T shared by 5% of the population. If Andrew has type T (event X), then

$$P(A \mid X) = \frac{P(A \cap X)}{P(X)} = \frac{P(X \mid A)P(A)}{P(X)}. \tag{15.3}$$

Here $P(X \mid A) = 1$, $P(X \mid B) = \frac{1}{20}$, $P(A) = P(B) = \frac{1}{2}$, so $P(X) = \frac{1}{2} + \frac{1}{40} = \frac{21}{40}$ by (15.2), and $P(A \mid X) = \frac{20}{21}$.

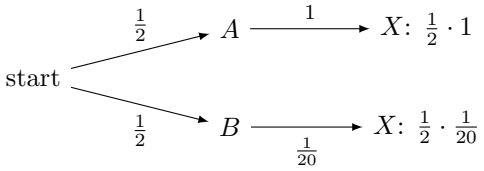


Figure. *Example 15.14 as a probability tree:* $P(A \mid X) = \frac{1/2}{1/2+1/40} = \frac{20}{21}$.

15.3.2 More than two events

Pairwise independence is not enough. Pick $x \in [10]$ and $y \in [20]$, and let $A = \{x \text{ odd}\}$, $B = \{y \text{ odd}\}$, $C = \{x - y \text{ odd}\}$. Each has probability $\frac{1}{2}$ and any two are independent, yet $P(A \cap B \cap C) = 0 \neq \frac{1}{8}$.

Definition 15.15 (Mutual independence)

$A_1, \dots, A_n$ are **independent** if $P(A_{i_1} \cap \cdots \cap A_{i_k}) = P(A_{i_1}) \cdots P(A_{i_k})$ for every nonempty $\{i_1, \dots, i_k\} \subseteq [n]$.

Theorem 15.16 (Bayes' theorem, general version)

If $A_1, \dots, A_n$ are pairwise disjoint with union Ω and C is any event, then $P(C) = \sum_{i=1}^{n} P(C \mid A_i)P(A_i)$ (terms with $P(A_i) = 0$ omitted).

Proof. C is the disjoint union of the sets $C \cap A_i$, and $P(C \cap A_i) = P(C \mid A_i)P(A_i)$. ∎

Monty Hall. The contestant picks door 1; the host, who knows where the car is, opens door 2, showing a goat (event B). With $A_i =$ "car behind door i",

$$P(A_1 \mid B) = \frac{P(B \mid A_1)P(A_1)}{\sum_i P(B \mid A_i)P(A_i)} = \frac{\frac{1}{2} \cdot \frac{1}{3}}{\frac{1}{2} \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3}} = \frac{1}{3}, \tag{15.4}$$

so switching wins with probability $\frac{2}{3}$.

15.4 Expected values

Motivation. Expectation is linear even for heavily dependent variables. So averages of complicated counts (valleys, fixed points) split into sums of easy indicator expectations. And since some outcome is at least the average, expectations also prove existence.

A **random variable** is a function $X : \Omega \to \mathbb{R}$; sums and products are taken pointwise.

Definition 15.17 (Expectation)

If X takes finitely many values S, its **expected value** is $E(X) = \sum_{i \in S} i\, P(X = i)$, where $P(X = i) = \frac{|\{u : X(u) = i\}|}{|\Omega|}$. Equivalently,

$$E(X) = \frac{1}{|\Omega|} \sum_{u \in \Omega} X(u). \tag{15.5}$$

The value depends on Ω (write E_Ω if needed); countable ranges need convergent sums.

Definition 15.18 (Independent variables)

X, Y are **independent** if $P(X = s, Y = t) = P(X = s)P(Y = t)$ for all s, t.

15.4.1 Linearity of expectation

Theorem 15.19 (Linearity)

(1) $E(X + Y) = E(X) + E(Y)$ for random variables on the same space.
(2) $E(cX) = cE(X)$ for $c \in \mathbb{R}$.

Proof.

Step 1 (equally likely outcomes). Both follow from (15.5): $\sum_u (X + Y)(u) = \sum_u X(u) + \sum_u Y(u)$ and $\sum_u cX(u) = c \sum_u X(u)$.

Step 2 (without uniformity). Let X take values x_i and Y values y_j. Then

$$E(X + Y) = \sum_{i,j} (x_i + y_j) P(X = x_i, Y = y_j) = \sum_i x_i P(X = x_i) + \sum_j y_j P(Y = y_j),$$

summing out j and i respectively. For (2), $P(cX = cx_i) = P(X = x_i)$ if $c \neq 0$ (and $c = 0$ is trivial). ■

No independence is needed. An **indicator variable** is 1 if an event occurs and 0 otherwise, so its expectation is the probability of the event.

i is a **valley** of p if $2 \leq i \leq n - 1$ and $p_i < p_{i-1}$, $p_i < p_{i+1}$.

Theorem 15.20

For $n \geq 2$, a random n-permutation has on average $\frac{n-2}{3}$ valleys.

Proof. Let Y_i $(2 \leq i \leq n - 1)$ indicate that i is a valley. Each of p_{i-1}, p_i, p_{i+1} is equally likely to be the smallest of the three, so $E(Y_i) = \frac{1}{3}$. The number of valleys is $Y = \sum_{i=2}^{n-1} Y_i$, so $E(Y) = \frac{n-2}{3}$ by Theorem 15.19. ■

Theorem 15.21

A random n-permutation has on average one fixed point.

Proof. Let X_i indicate $p_i = i$; p_i is uniform on $[n]$, so $E(X_i) = \frac{1}{n}$. Then

$$E(X) = \sum_{i=1}^{n} E(X_i) = n \cdot \frac{1}{n} = 1. \tag{15.6}$$

■

15.4.2 Existence proofs using expectation

Theorem 15.22

If X takes finitely many values with maximum j, then $j \geq E(X)$. Hence some outcome u has $X(u) \geq E(X)$.

Proof. $E(X) = \sum_{i \in S} i\, P(X = i) \leq j \sum_{i \in S} P(X = i) = j$. ■

Theorem 15.23

A simple graph G on $[n]$ ($n \geq 2$) with $m \geq 1$ edges has a bipartite subgraph with more than $m/2$ edges.

Proof. Let Ω be the $2^{n-1} - 1$ splits of $[n]$ into two nonempty parts $\{A, B\}$; each split gives the bipartite subgraph of edges between A and B, with X edges. Let X_i indicate that edge i crosses. The splits separating its endpoints u, v are obtained by sending each of the other $n - 2$ vertices to u's or v's side, so

$$E(X_i) = P(X_i = 1) = \frac{2^{n-2}}{2^{n-1} - 1} > \frac{1}{2}.$$

Then $E(X) = \sum_{i=1}^{m} E(X_i) > \frac{m}{2}$, and Theorem 15.22 gives a split with more than $m/2$ crossing edges. ■

Example 15.24 (Betweenness)

Given triples $L_i = (a_i, b_i, c_i)$ of distinct elements of $[n]$, $i \leq k$, say that p satisfies L_i if b_i lies between a_i and c_i in p. Some n-permutation satisfies at least $k/3$ of the triples.
Solution. For a random p, each of a_i, b_i, c_i is equally likely to be in the middle, so the indicator Y_i of satisfying L_i has $E(Y_i) = \frac{1}{3}$. The number of satisfied triples has expectation $\frac{k}{3}$; apply Theorem 15.22.

15.4.3 Conditional expectation

$E(X \mid A) = \sum_i i\, P(X = i \mid A)$ is the expectation of X given A ($P(A) > 0$).

Theorem 15.25

If $A_1, \ldots, A_n$ are pairwise disjoint with union Ω (and positive probabilities), then $E(X) = \sum_{i=1}^{n} E(X \mid A_i) P(A_i)$.

Proof. By Theorem 15.16 with $C = \{X = j\}$, $P(X = j) = \sum_i P(X = j \mid A_i) P(A_i)$. Multiply by j and sum over j:

$$E(X) = \sum_j j \sum_i P(X = j \mid A_i) P(A_i) = \sum_i P(A_i) \sum_j j\, P(X = j \mid A_i) = \sum_i E(X \mid A_i) P(A_i).$$

■

Example 15.26

A die is thrown three times and the first throw is at least 4. The first throw is even (A_1) with probability $\frac{2}{3}$, giving on average $1 + 2 \cdot \frac{1}{2} = 2$ even throws, and odd (A_2) with probability $\frac{1}{3}$, giving on average 1. So the expected number of even throws is $\frac{2}{3} \cdot 2 + \frac{1}{3} \cdot 1 = \frac{5}{3}$.

Example 15.27

A team wins each game independently with probability $\frac{3}{4}$. Given at least three wins in the first four games (B), what is the expected number of wins in 12 games?
Solution. Let A_1, A_2 be "exactly 3" and "all 4" of the first four won. $P(A_1) = 4 \cdot \frac{1}{4}(\frac{3}{4})^3 = \frac{27}{64}$ and $P(A_2) = (\frac{3}{4})^4 = \frac{81}{256}$, so $P_B(A_1) = \frac{4}{7}$ and $P_B(A_2) = \frac{3}{7}$. The last eight games add $8 \cdot \frac{3}{4} = 6$ wins on average. By Theorem 15.25 in the space B,

$$E_B(X) = 9 \cdot \frac{4}{7} + 10 \cdot \frac{3}{7} = \frac{66}{7},$$

more than the unconditional 9.

15.5 Selected exercises

Motivation. The two book exercises below are the typical uses of indicator variables: an existence proof from an expectation, and a second-moment computation that needs the joint behaviour of pairs of indicators.

Exercise 15.17 (Tournaments with many Hamiltonian paths)

Prove that for every $n \geq 1$ there is a tournament on n vertices with at least $\frac{n!}{2^{n-1}}$ Hamiltonian paths. What can be said about Hamiltonian cycles?

Solution.

Step 1 (random tournament). Orient each edge of K_n on $[n]$ independently in either direction with probability $\frac{1}{2}$; all $2^{\binom{n}{2}}$ tournaments are equally likely.

Step 2 (indicators). For a permutation $\sigma = \sigma_1 \cdots \sigma_n$ of $[n]$ let X_σ be the indicator of the event that $\sigma_1 \to \sigma_2 \to \cdots \to \sigma_n$ is a directed path. This event prescribes the directions of $n-1$ distinct edges, so $E(X_\sigma) = 2^{-(n-1)}$.

Step 3 (expectation). The number of Hamiltonian paths is $X = \sum_\sigma X_\sigma$, so by Theorem 15.19, $E(X) = \frac{n!}{2^{n-1}}$. By Theorem 15.22, some tournament has $X \geq \frac{n!}{2^{n-1}}$.

For cycles ($n \geq 3$): a directed Hamiltonian cycle is determined by a cyclic order of $[n]$, and there are $(n-1)!$ of them (Example 6.10). Each prescribes the directions of n distinct edges, so it occurs with probability 2^{-n}. The same argument gives a tournament with at least $\frac{(n-1)!}{2^n}$ Hamiltonian cycles. ■

Exercise 15.18 (The variance of the number of fixed points)

For a random variable Y let $\mathrm{Var}(Y) = E((Y - E(Y))^2)$.
(a) Prove that $\mathrm{Var}(Y) = E(Y^2) - E(Y)^2$.
(b) Let $X(p)$ be the number of fixed points of a random n-permutation p, $n \geq 2$. Prove that $\mathrm{Var}(X) = 1$.

Solution.

(a) Let $\mu = E(Y)$, a constant. By Theorem 15.19,

$$E((Y-\mu)^2) = E(Y^2) - 2\mu E(Y) + \mu^2 = E(Y^2) - \mu^2.$$

(b)

Step 1 (indicators). Let X_i be the indicator of $p(i) = i$, so $X = \sum_i X_i$ and, as in Theorem 15.21, $E(X_i) = \frac{(n-1)!}{n!} = \frac{1}{n}$ and $E(X) = 1$.

Step 2 (second moment). Since $X_i^2 = X_i$,

$$X^2 = \sum_i X_i + \sum_{i \neq j} X_i X_j.$$

For $i \neq j$, $X_i X_j$ is the indicator of $p(i) = i$ and $p(j) = j$; exactly $(n-2)!$ permutations fix both, so $E(X_i X_j) = \frac{(n-2)!}{n!} = \frac{1}{n(n-1)}$. There are $n(n-1)$ ordered pairs $i \neq j$, hence

$$E(X^2) = n \cdot \frac{1}{n} + n(n-1) \cdot \frac{1}{n(n-1)} = 2.$$

Step 3 (conclusion). By (a), $\mathrm{Var}(X) = 2 - 1^2 = 1$. (For $n = 1$, $X \equiv 1$ and the variance is 0.)

So the number of fixed points has mean 1 and variance 1 for every $n \geq 2$, like a Poisson variable with parameter 1. ■